# HAOWEN (JOHN) WEI

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## EDUCATION

**Columbia University** 09/2022 - 05/2024

Master of Computer Science (Thesis Track), GPA: 3.81/4.00

- Advisors: Prof. Steven K. Feiner & Prof. Paul Sajda & Prof. Kaveri Thakoor
- Master's Thesis: From Brain–Computer Interfaces to AI-Enhanced Diagnostics: Developing Cutting-Edge Tools for Medical and Interactive Technologies. Paper

**Worcester Polytechnic Institute** 08/2018 - 05/2022

Bachelor of Computer Science & Electrical and Computer Engineering (Double Major), GPA: 3.91/4.00

- Advisors: Prof. Erin Solovey & Prof. Kaveh Pahlavan
- Dean's List: Fall 2018, Spring 2019, Fall 2019, Spring 2021, Fall 2021; Graduated with Distinction.
- 2022 Best Undergraduate Major Qualifying Project (MQP), 3rd Place (IndexPen)

## RESEARCH INTEREST

Ultrasound Imaging, Multimodal AI, Deep Learning, Machine Learning, Computer Vision, Signal Processing, Human-Computer Interaction, Virtual Reality, Augmented Reality, Eye-Tracking, Brain-Computer Interface, Wireless Sensing.

## SELECTED PUBLICATIONS

*** These Authors Contributed Equally.**

- David Dai, Jeannie She, Jiaee Cheong, Xing Han, Carl Harris, **Haowen Wei**, Farzan Vahedifard, Suchi Saria, Robert Stevens, Paul Pu Liang. "QoQ-Med3: A Multimodal Reasoning Foundation Model for Clinical Analysis." npj Digital Medicine (2026). Paper
- Ziheng Li*, **Haowen Wei***, Kuang Sun, Leyi Cui, David Li, Steven Feiner, Kaveri Thakoor. "Interactively Assisting Glaucoma Diagnosis with an Expert Knowledge-Distilled Vision Transformer." ACM CHI Conference on Human Factors in Computing Systems Late-Breaking Work 2025. Paper, Video
- Wei Dai, Peilin Chen, Malinda Lu, Daniel Li, **Haowen Wei**, Hejie Cui, and Paul Pu Liang. "CLIMB: Data Foundations for Large Scale Multimodal Clinical Foundation Models." ICML 2025. Paper
- Ziheng Li*, **Haowen Wei***, Ziwen Xie, Yunxiang Peng, June Pyo Suh, Steven Feiner, Paul Sajda. "PhysioLabXR: A Python Platform for Real-Time, Multi-modal, Brain–Computer Interfaces and Extended Reality Experiments." Journal of Open Source Software 9, no. 93 (2024): 5854. Paper, GitHub
- **Haowen Wei***, Ziheng Li*, Alexander D. Galvan, Zhuoran Su, Xiao Zhang, Kaveh Pahlavan, and Erin T. Solovey. "IndexPen: Two-Finger Text Input with Millimeter-Wave Radar." Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies 6, no. 2 (2022): 1-39. Paper, Video, Dataset
- Xing Xie, **Haowen Wei**, and Yongjie Yang. "Real-Time LiDAR Point-Cloud Moving Object Segmentation for Autonomous Driving." Sensors 23, no. 1 (2023): 547. Paper
- Ziheng Li*, Xichen He*, Mengyuan Wu, Zeyi Tong, **Haowen Wei**, Benjamin Yang, Steven Feiner, Paul Sajda. "SwEYEpinch and Beyond: Exploring Intuitive, Efficient Text Entry for Extended Reality via Eye and Hand Tracking." ACM CHI Conference on Human Factors in Computing Systems (CHI 2026). Paper

## CONFERENCE PRESENTATIONS

- 2023 Brain & Human Body Modeling (BHBM) - Harvard Medical School Martinos Bioimaging Center (Hybrid Local Conference): PhysioLabXR: A software platform in Python for multi-modal brain-computer interface and real-time experiment pipelines. **Haowen Wei**, Ziheng Li, Steven Feiner, Paul Sajda Link
- 2022 Brain & Human Body Modeling (BHBM) - Harvard Medical School Martinos Bioimaging Center (Hybrid Local Conference): Hardware, real-time signal processing techniques, and data collection for TMS induced EMG responses with RenaLabApp. **Haowen Wei**, Mohammad Daneshzand Link
- 2021 Brain & Human Body Modeling (BHBM) - Harvard Medical School Martinos Bioimaging Center (Hybrid Local Conference): Interplay between TES and EEG with boundary element fast multipole method (BEM-FMM) via Helmholtz reciprocity principle. Sergey Makarov, **Haowen Wei**, Aapo Nummenmaa Link

## SELECTED RESEARCH PROJECTS

**UltraPose: Simultaneous Hand Pose Reconstruction and Finger Force Estimation from Forearm Ultrasound**

Advisor: Dr. Brian Anthony — 07/2026 - Present
Role: Project Lead — *Massachusetts Institute of Technology*

- Founded and led a project that reconstructs 20-DOF hand pose and estimates finger force simultaneously from a single forearm ultrasound probe, imaging the muscles that drive the hand rather than instrumenting the hand itself.
- Designed and built a multimodal acquisition platform providing time-synchronized capture of Telemed forearm ultrasound, Manus glove joint angles, video, and force-sensitive resistor (FSR) pinch force.
- Developed the processing pipeline and multi-task deep models that jointly regress finger joint angles and pinch force, using the glove and FSR streams as ground truth.

**Piano Biomechanics: Multimodal Sensing of Tissue Dynamics in Expert Piano Performance** [MIT News]

Advisors: Dr. Praneeth Namburi & Dr. Mi-Eun Kim — 07/2025 - Present
Role: Research Associate & Research Technician — *Massachusetts Institute of Technology*

- Led 30 five-hour human-subjects sessions in a study now at 65 of 80 planned participants, capturing time-synchronized dual-arm ultrasound, surface EMG, optical motion capture, and key-force data from elite and student pianists during performance.
- Led cross-modal synchronization analysis across Yamaha Disklavier, Telemed ultrasound, Delsys EMG, and OptiTrack motion capture, achieving 3 ms alignment across all four streams.
- Led development of the lab's multimodal annotation toolchain, enabling frame-accurate labeling and joint review of all four modalities in a single interface.
- Built a model-assisted point-tracking pipeline spanning 200+ hours of ultrasound video, hand-labeling a subset to train a model that propagates annotations across the remainder, to quantify how connective and elastic tissue motion differs across expertise levels.

**PhysioLabXR: A Python Platform for Real-Time, Multi-modal, Brain–Computer Interface and Extended Reality Experiments** [GitHub]

Advisors: Prof. Steven K. Feiner & Prof. Paul Sajda — 05/2020 - 10/2025
Role: Project Lead — *WPI & Columbia University*

- Co-founded and led development of an open-source platform for real-time acquisition, visualization, and neural network inference over heterogeneous physiological and XR data streams; published in the Journal of Open Source Software.
- Architected the data visualization, user interface, and digital signal processing subsystems, including a real-time TMS/fMRI visualizer and serial interfaces for sensors outside the Lab Streaming Layer ecosystem.
- Built reference Brain-Computer Interface paradigms (e.g., P300 speller) and maintained the platform for research groups at WPI and Columbia University.

**Interactively Assisting Glaucoma Diagnosis with an Expert Knowledge-Distilled Vision Transformer**
Advisors: Prof. Steven K. Feiner & Prof. Kaveri Thakoor 04/2023 - 07/2025
Role: Project Lead *Columbia University*

- Developed Interactive Area-of-Interest (AOI) Augmentation, which surfaces a vision transformer's attention over ocular images to direct clinician focus during diagnosis.
- Led end-to-end system development in Unity and PhysioLabXR, covering architecture, interaction design, and real-time rendering during diagnostic review.
- Ran two eye-tracking user studies with 13 clinicians (ophthalmology residents, glaucoma specialists, and an OCT technician): interactive perceptual guidance improved diagnostic accuracy over static guidance (90% vs. 88%), and ViT guidance reduced variance in report-viewing behavior without reducing accuracy.

**IndexPen: Two-Finger Text Input with Millimeter-Wave Radar**
Advisors: Prof. Erin Solovey & Prof. Kaveh Pahlavan 08/2019 - 09/2022
Role: Project Lead *Worcester Polytechnic Institute*

- Introduced a touch-free, in-air text entry system that recognizes a 30-character set from two-finger micro-gestures using mmWave radar, with no instrumentation on the hand.
- Led the work end to end: radio frequency processing pipeline, CNN+LSTM recognition architecture, human-subjects data collection, and real-time inference.
- Achieved 95.89% 10-fold cross-validation accuracy across 31 classes over 31,000 gesture samples, and validated learnability with 16 first-time users across five sessions.

## RESEARCH EXPERIENCE

**Research Associate & Research Technician**, Massachusetts Institute of Technology 07/2025 - Present
MIT.nano Immersion Lab & Device Realization Laboratory

- Design and run human-subjects studies on ultrasound-based musculoskeletal sensing, spanning instrumentation, protocol, data collection, and modeling.
- Maintain the lab's multimodal capture stack (ultrasound, EMG, motion capture, force) for ongoing studies and prepare manuscripts for publication.

**Research Assistant**, Harvard Medical School 12/2021 - 09/2022, 09/2024 - 06/2025
Athinoula A. Martinos Center for Biomedical Imaging (TMS Lab)

- Built a real-time TMS-EEG signal processing pipeline for artifact handling and brain signal reconstruction.
- Led development of a unified device API for EEG, TMS, and EMG hardware, substantially reducing experiment setup time for the research team.

**Research Assistant**, Columbia University 09/2022 - 05/2024
LIINC Lab, CGUI Lab, & AI4VS Lab

- Led PhysioLabXR development from scratch and built multimodal physiological classifiers and Unity experiment paradigms used across three labs.
- Work from this period appears in CHI 2025, CHI 2026, and the Journal of Open Source Software, alongside a thesis-track master's degree.

**Research Assistant**, Worcester Polytechnic Institute 10/2019 - 05/2022
CWINS Lab, HCI Lab, & Embedded Systems Lab

- Led the IndexPen mmWave radar gesture sensing project and contributed to four publications in wireless sensing and human-computer interaction venues.
- Supported other research teams with software and hardware, spanning mmWave radar setup, UWB and BLE proximity estimation, and LiDAR point cloud moving object segmentation.

## TEACHING & MENTORING

**Undergraduate Research Mentoring**, Massachusetts Institute of Technology 07/2025 - Present
MIT.nano Immersion Lab

- Train and supervise undergraduate researchers on experimental protocol, multimodal hardware setup, and the ultrasound annotation tooling used in the piano biomechanics study.

**Teaching Assistant**, Worcester Polytechnic Institute 10/2020 - 12/2021
ECE 2029: Introduction to Digital Circuit Design; ECE 3308: Introduction to Wireless Networks

- Supported students through homework and coding problems, guided MATLAB and Verilog implementations, and graded exams and assignments.
- Helped students structure project workflows to meet lab objectives.

## SKILLS

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|---|---|
| **Programming** | Python, C#, C/C++, Java, MATLAB, Verilog/FPGA, SQL, JavaScript, HTML |
| **Research** | Multimodal AI, Deep Learning, Machine Learning, Computer Vision, Image Processing, Real-Time Signal Processing, Sensor Fusion & Multimodal Synchronization, Ultrasound Imaging, Motion Capture, Electromyography (EMG), Physiological Data Processing (EEG, fNIRS), Brain Stimulation (TMS), Eye Tracking, mmWave Radar, UWB Radar, LiDAR Point Cloud Segmentation, Natural Language Processing, Human-Subjects Study Design |
| **Platform** | PyTorch, TensorFlow, Unity3D, Qt/PyQt, Lab Streaming Layer, Telemed Ultrasound, OptiTrack, Delsys EMG, Manus Glove, SOLIDWORKS, FPGA, Real-Time Operating System (RTOS) |